

Lecture 2.1: The Language of Motion

Unit 2: The Art of Observation

Today's Mission

- **Bridge Back:** In the rocket lab, the changing slope of the velocity graph told a story about the engine firing and cutting off. Today, we'll formalize the rules for reading that story.
 - **Thinking Lens:** We will look for **Patterns** in graphs to understand the story of an object's motion.
 - **Essential Questions:**
 - 1. What is the difference between distance and displacement?
 - 2. How can we find velocity from a position-time graph?
 - 3. What do the slope and area of a velocity-time graph represent?
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Where Are You? Position, Distance, & Displacement

- **Position (\$x\$):** An object's location relative to a defined origin.
 - **Distance:** The total path length traveled. It is a **scalar**.
 - **Displacement (\$\Delta \vec{x}\$):** The change in position (\$\vec{x}_{final} - \vec{x}_{initial}\$). It is a **vector** (magnitude and direction).
 - **Example:** You walk \$7.0 \text{ m}\$ East, then turn around and walk \$3.0 \text{ m}\$ West.
 - **Distance traveled:** \$7.0 \text{ m} + 3.0 \text{ m} = 10.0 \text{ m}\$
 - **Displacement:** \$4.0 \text{ m, East}\$
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Position vs. Time Graphs

A position vs. time graph shows an object's position at every moment.

The most important rule: The **slope** of a position vs. time graph represents the object's **velocity**.

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{\Delta x}{\Delta t} = \text{velocity}$$

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Position vs. Time: Accelerated Motion

When an object accelerates, its position vs. time graph becomes curved.

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Reading the Story of a P-T Graph

- **Positive Slope:** Object is moving in the positive direction (away from the origin).
 - **Negative Slope:** Object is moving in the negative direction (toward the origin).
 - **Zero Slope (Horizontal Line):** Object is at rest (velocity is zero).
 - **Steeper Slope:** Higher speed.
 - **Curved Line:** The velocity is changing (acceleration is occurring).
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Velocity vs. Time Graphs

1. The **slope** of the graph represents **acceleration**. $\text{slope} = \frac{\Delta v}{\Delta t} = \text{acceleration}$
2. The **area** under the graph represents **displacement**. $\text{area} = \text{velocity} \times \text{time} = \frac{1}{2} \times \text{base} \times \text{height} = \text{displacement}$

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Velocity vs. Time: Accelerated Motion

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Key observations:

- Constant slope = constant acceleration
 - Area under the curve = total displacement
 - Steeper slope = greater acceleration
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Example: Analyzing the Rocket Launch

Let's analyze the first 10 seconds of the rocket's motion from our lab data.

- **Acceleration (\$0-4s\$):** $\text{slope} = \frac{40.0 \text{ m/s} - 0.0 \text{ m/s}}{4.0 \text{ s} - 0.0 \text{ s}} = 10.0 \text{ m/s}^2$
 - **Displacement (\$0-4s\$):** $\text{area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} (4.0 \text{ s})(40.0 \text{ m/s}) = 80.0 \text{ m}$
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Check for Understanding

Describe the rocket's motion from **t = 4.0 s to t = 8.2 s**.

1. **What is its acceleration?**
 - It's the slope. It's constant and negative ($\approx -9.8 \text{ m/s}^2$).
2. **Is it speeding up or slowing down?**
 - It's slowing down. The velocity is positive but decreasing.
3. **How could we find its displacement during this time?**
 - Find the area of the trapezoid between t=4s and t=8.2s.

Answering Our Essential Questions

1. What is the difference between distance and displacement?

- **Distance** is the total path length traveled (scalar)
- **Displacement** is the change in position with direction (vector)

2. How can we find velocity from a position-time graph?

- **Velocity** = **slope** of the position-time graph
- Steeper slope = higher speed; negative slope = motion toward origin

3. What do the slope and area of a velocity-time graph represent?

- **Slope** = acceleration (rate of velocity change)
 - **Area** = displacement (total change in position)
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Summary and Next Steps

Key Rules:

- **Position-Time:**
 - Slope = Velocity
- **Velocity-Time:**
 - Slope = Acceleration
 - Area = Displacement
- **Bookend Statement:** A graph of motion is a story waiting to be read.
- **Bridge Forward:** The skills of interpreting slope and area on these motion graphs are exactly what you'll need to master **Problem Set 2.1**.